

SOLDOTNA, AK, USAWMO: **702595**Lat: **60.476N**Lon: **151.034W**Elev: **113**StdP: **14.64**Time Zone: **-9.00 (NAK)**Period: **94-19**WBAN: **26559****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.0	-16.9	-27.3	1.2	-16.9	-24.3	1.4	-11.9	16.1	34.0	13.2	32.2	2.3	100	0.684

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.6	72.4	58.7	69.5	57.2	65.8	55.2	60.0	70.1	58.1	66.5	56.4	63.9	7.6	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.9	64.4	63.2	53.9	62.1	61.7	52.3	58.5	59.9	26.4	70.5	25.1	66.1	24.0	63.8	68.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.1	12.8	11.2	-23.6	76.6	7.6	4.2	-29.1	79.6	-33.6	82.0	-37.9	84.4	-43.4	87.4		
(5)			-23.1	61.6	6.5	2.7	-27.8	63.5	-31.6	65.1	-35.3	66.7	-40.0	68.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	35.2	13.3	19.8	23.8	36.0	44.9	52.0	56.2	54.9	47.6	35.1	21.8	16.5
	DBStd	17.75	15.50	12.05	11.02	6.56	4.80	4.29	3.53	4.01	5.01	8.75	11.55	13.30
	HDD50	5865	1139	846	813	421	168	25	1	5	99	462	846	1040
	HDD65	10869	1604	1266	1278	871	624	391	273	314	521	926	1296	1505
	CDD50	475	0	0	0	0	10	84	194	158	28	1	0	0
	CDD65	2	0	0	0	0	0	0	1	1	0	0	0	0
	CDH74	47	0	0	0	0	0	11	24	12	0	0	0	0
	CDH80	4	0	0	0	0	0	0	3	1	0	0	0	0
Wind	WSAvg	3.9	3.3	3.3	3.8	4.5	5.4	4.9	4.2	3.8	3.6	3.5	3.4	3.5
Precipitation	PrecAvg	23.4	1.6	1.5	1.0	0.9	0.9	1.3	2.2	2.7	3.7	3.1	2.2	2.2
	PrecMax	31.0	3.7	3.8	3.0	2.6	2.5	2.6	4.5	5.0	6.6	8.1	4.9	4.6
	PrecMin	18.3	0.3	0.0	0.2	0.2	0.2	0.4	0.7	0.5	1.4	1.7	0.0	0.4
	PrecStd	3.1	0.9	0.7	0.7	0.6	0.6	0.6	1.1	1.1	1.5	1.5	1.5	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.2	43.1	47.8	58.6	70.2	75.4	77.1	75.6	65.8	56.6	46.3	42.7
		MCWB	38.3	37.0	39.4	45.6	52.2	58.7	60.7	62.1	53.3	48.5	40.9	38.5
	2%	DB	38.8	39.2	44.7	53.9	64.0	70.8	72.4	71.3	62.5	52.1	42.7	38.6
		MCWB	35.1	34.9	37.3	42.4	49.5	56.1	59.6	59.6	52.1	45.5	38.6	35.7
	5%	DB	36.4	37.0	41.1	50.7	60.8	66.2	69.0	66.5	59.2	49.3	39.0	36.2
		MCWB	33.3	33.3	35.0	40.8	47.5	54.2	58.1	56.3	50.5	44.1	36.0	33.9
	10%	DB	33.9	35.7	38.6	47.7	56.7	63.4	65.5	64.1	56.7	46.4	36.7	33.8
		MCWB	31.6	32.6	33.1	39.3	46.0	52.5	56.2	54.9	49.7	42.3	34.4	32.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.8	37.9	40.7	46.8	53.5	60.5	62.4	63.2	55.4	49.9	41.8	39.0
		MCDB	42.8	41.5	46.1	57.0	66.5	73.7	73.7	73.8	62.2	55.0	45.4	41.9
	2%	WB	35.5	35.9	38.0	43.1	50.4	57.1	60.0	60.2	53.5	46.7	39.3	36.1
		MCDB	38.1	39.1	43.3	51.5	62.0	67.8	70.1	69.4	58.8	50.1	42.2	38.3
	5%	WB	33.5	34.0	35.5	41.3	48.5	55.0	58.4	57.8	52.1	45.2	36.9	33.7
		MCDB	35.9	36.7	40.5	48.9	58.5	64.7	67.0	65.3	57.4	48.4	39.0	35.6
	10%	WB	31.3	32.3	33.7	39.6	46.8	53.1	56.9	56.0	50.8	43.2	34.6	31.7
		MCDB	33.5	34.8	38.1	47.0	55.7	62.1	64.3	62.5	55.8	46.4	36.6	33.2
Mean Daily Temperature Range	5% DB	MDBR	14.3	17.4	20.6	17.9	19.1	17.8	15.6	16.4	15.9	14.8	13.4	13.4
		MCDBR	12.8	14.3	18.1	22.3	28.6	26.8	23.7	23.1	20.3	16.2	11.2	12.6
		MCWBR	10.5	11.3	12.8	13.7	16.5	14.9	13.2	13.3	12.6	10.9	8.9	10.6
	5% WB	MCDBR	12.5	13.2	16.3	19.7	24.0	23.2	20.3	19.2	16.3	13.3	10.9	12.0
		MCWBR	10.5	10.6	11.8	12.6	14.5	13.6	12.0	11.8	10.9	9.4	8.8	10.6
Clear-Sky Solar Irradiance	taub		0.239	0.283	0.301	0.345	0.372	0.362	0.369	0.353	0.326	0.300	0.252	0.226
	taud		2.163	2.285	2.322	2.323	2.327	2.437	2.448	2.495	2.551	2.497	2.171	1.980
	Ebn at Noon		170	228	264	270	270	276	270	266	256	225	158	110
	Edh at Noon		12	20	27	33	36	32	31	28	22	17	12	9
All-Sky Solar Radiation	RadAvg		95	306	792	1274	1648	1724	1520	1209	784	373	136	48
	RadStd		11	42	92	129	131	124	147	144	111	39	23	7